
pythia: a Lean 4 tactic library for statistical proof automation

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Abstract

Lean 4 with Mathlib ships strong general automation (`aesop`, `simp`, `linarith`, `polyrith`, `nlinarith`, `positivity`, `measurability`, `bound`, `gcongr`). It does not ship a hammer for applied mathematics: the moment a goal mentions a martingale, a stopping time, a production function, an ODE Lyapunov candidate, a Wasserstein distance, or a quantum density operator, the generic tactics have nothing to apply. We present *pythia*, a Lean 4 tactic library for applied-mathematics proof automation. The headline tactic `pythia!` orchestrates a nine-rung deterministic ladder (`stat_simp`, `linarith` / `nlinarith` / `polyrith`, `positivity`, `aesop` on the registered *Pythia* ruleset, the `@[stat_lemma]` cascade, `z3_check`, `cvc5_check`, `vampire_check` / `e_check`, and `disprove`); each rung tries fail-fast with a bounded budget; the verbose variant `pythia?` reports the closing rung and per-rung wall-clock timing for diagnostic use. The library ships ten registered tactics, attribute-driven registries (`@[stat_lemma]`, `@[stats_ineq]`, `@[prob_simp]`, `@[stat_simp]`, `@[anytime_valid_lemma]`, plus the per-domain calculator typeclass), and a fifteen-domain proof library with thirty-two cross-domain theorems carrying matched Python `sim` runners (property-based testing, deterministic sweeps, mutation testing). Domains span probability and statistics, biology, chemistry, economics, control theory, optimal transport, quantum information, stochastic calculus, game theory, thermodynamics, numerical analysis, queueing, and operations research. Every public

theorem is axiom-clean against the standard Lean kernel set (`propext`, `Classical.choice`, `Quot.sound`). The cross-prover ladder is offline-first and LLM-free: `z3_check` / `cvc5_check` / `vampire_check` / `e_check` dispatch to local SMT and ATP binaries with kernel-checked reconstruction in the CoqHammer style of Czajka & Kaliszyk (2018); LLM-augmented closure lives in a separate companion SDK that the library does not depend on. Hard structural Mathlib gaps are optionally routed to the Aristotle automated theorem prover (Harmonic, 2025) for development-time restocking, with audit gates against vacuous-truth closures. The library is pinned to Lean 4.28 + Mathlib v4.28.0 and released open-source.

1. Introduction

Lean 4 with Mathlib ships strong general-purpose proof automation: `aesop` for forward search, `simp` for rewriting, `linarith` and `polyrith` for linear and polynomial arithmetic, `nlinarith` for non-linear arithmetic, `positivity` for sign goals, `measurability` for the σ -algebra fragment, `bound` for monotone composition, `gcongr` for generalised congruence (de Moura & Ullrich, 2021; The Mathlib Community, 2020; 2025). Applied-mathematics proofs route through these general tactics, typically by hand decomposition into a chain of `simp` sets, `linarith` hypotheses, and Mathlib lemma selection. None of the general tactics know about each other; the user is the orchestrator. There is no domain hammer for applied mathematics.

CoqHammer (Czajka & Kaliszyk, 2018) dispatches Coq goals to external ATPs (Vampire, E, CVC4) with kernel-checked reconstruction. Sledgehammer (Paulson & Blanchette, 2010; Blanchette et al., 2011) plays the same role in Isabelle/HOL. Lean 4 has no comparable hammer. This paper does not attempt the fully-general case across all of Lean. We address the applied-mathematics surface: a registered library of concentration, anytime-valid, optimal transport, control-theoretic, and information-theoretic

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lemmas plus a fan-out across local SMT and ATP solvers, packaged as a single deterministic tactic call.

Contribution. We present *pythia*, a Lean 4 tactic library for applied-mathematics proof automation. The headline tactic *pythia!* orchestrates a nine-rung deterministic ladder under one call: *stat.simp* (curated probability and ENNReal normal forms), *linarith* / *nlinarith* / *polyarith*, *positivity*, *aesop* on the registered *Pythia* ruleset, the *@[stat.lemma]* shape-dispatch cascade, *z3.check* (QFLRA via Z3), *cvc5.check* (QF.BV / QFLRA via CVC5), *vampire.check* / *e.check* (FOL via Vampire / E), and *disprove* (counterexample finder via Z3 model extraction). Each rung tries fail-fast with a per-rung budget; first-to-close wins. The verbose variant *pythia?* reports the closing rung and per-rung wall-clock timing for diagnostic use. *pythia* ships ten registered tactics in total, six attribute-driven registries (*@[stat.lemma]*, *@[stats.ineq]*, *@[prob.simp]*, *@[stat.simp]*, *@[anytime.valid.lemma]*, plus the *DomainCalculator* typeclass for per-domain numerical calculators), and a fifteen-domain proof library with thirty-two cross-domain theorems carrying matched Python sim runners (property-based testing, deterministic sweeps, mutation testing). The shipped library is axiom-clean against *{propext, Classical.choice, Quot.sound}* and pinned to Lean 4.28 + Mathlib v4.28.0. The cross-prover ladder is offline-first and LLM-free: SMT and ATP backends are deterministic decision procedures with verifiable Lean reconstruction; LLM-augmented closure lives in a separate companion SDK that the library does not depend on. Hard structural Mathlib gaps are optionally routed to the Aristotle automated theorem prover (Harmonic, 2025) for development-time restocking only, with kernel-checked reconstruction inside Lean and an audit gate against vacuous closures.

Case study (vacuous-truth catch). On 2026-04-26 the Aristotle prover closed a candidate theorem in our *wald.wolfowitz.optimal* development by routing through a contradictory premise: a *Bool* partition combined with a log-boundary constraint forced both $\alpha \geq 1$ and $\alpha < 1$, closing the goal vacuously. The closure was kernel-checked. The theorem statement was the bug. The library audit caught the issue and forced a restatement to the two-measure form. We treat this kind of catch as a first-class library responsibility, not as a separate review pass.

Paper organisation. Section 2 describes the ten-tactic surface and the registry attributes. Section 3 groups the shipped library content by domain. Section 4 describes the Aristotle integration as a development-time oracle and the

vacuous-truth refusal policy. Section 5 compares against *aesop*, *Sledgehammer*, *CoqHammer*, and *Mathlib*’s existing tactic suite, states limitations, and reports a small Aristotle vs DSPv2 vs Opus 4.6 head-to-head on a subset of FormalAVS (Anonymous, 2026). Full numbers are in the dataset paper.

Reproduction and code. The *pythia* library is open source. The URL is redacted for double-blind review. Axiom audit transcripts and reproduction recipes are listed in Appendix D. The dataset of theorem-proving benchmarks (FormalAVS, Anonymous (2026)) is reported in a separate paper.

2. The *pythia* tactic surface

pythia ships ten user-facing tactics, one named *aesop* ruleset, and six attribute-driven registries. The design follows the *CoqHammer* pattern (Czajka & Kaliszyk, 2018): a domain-specific dispatcher with extensible attribute-driven lemma selection, a fall-through to general-purpose automation, and kernel-checked reconstruction for any external-oracle steps. The headline tactic *pythia!* orchestrates the full deterministic ladder under one call; the other tactics are exposed for direct invocation when the user knows the goal shape.

2.1. Ten tactics

***pythia* (top-level dispatcher).** The default entry point. Runs *aesop* with the *Pythia* *aesop* ruleset, which contains every theorem tagged *@[stat.lemma]*. On *aesop* failure, falls through to a short cleanup chain combining *simp*, *omega*, *linarith*, and *positivity*. The fall-through ordering is fixed. The lemma database and *aesop* ruleset are user-extensible.

***stats.ineq*.** Discharges sub-Gaussian, sub-gamma, Bernstein-Bennett, and Hoeffding-style inequalities. The *@[stats.ineq]* attribute re-tags a lemma with *Mathlib*’s *@[bound]*, so the *stats.ineq* tactic dispatches via *Mathlib*’s bound infrastructure first, then *positivity*, *gcongr*, *linarith*, and *nlinarith* in that order. Reconstruction is internal Lean and axiom-clean.

***prob.simp*.** Probability-domain rewrite tactic. The *@[prob.simp]* attribute forwards lemmas to *@[simp]* and additionally records the declaration name in a scoped extension for introspection. The tactic runs *simp* against the resulting set with a *push.cast* / *norm.cast* coercion fallback for ENNReal / NNReal / Real bridge. The *pdf.simp* syntax is a literal alias. Both forms expand to the same kernel trace.

Tactic	Role	z3_check.
pythia!	Headline orchestrator. Walks with linearith reconstruction inside Lean. Z3 supplies an unsat verdict. linearith reconstructs the proof term inside the kernel. The kernel is the final word. If Z3 says unsat but linearith cannot reconstruct, the tactic fails loud rather than silently trusting Z3. Z3 cannot smuggle axioms. This follows the CoqHammer reconstruction pattern (Czajka & Kaliszyk, 2018).	External Z3 oracle for linear real arithmetic, with linearith reconstruction inside Lean. Z3 supplies an unsat verdict. linearith reconstructs the proof term inside the kernel. The kernel is the final word. If Z3 says unsat but linearith cannot reconstruct, the tactic fails loud rather than silently trusting Z3. Z3 cannot smuggle axioms. This follows the CoqHammer reconstruction pattern (Czajka & Kaliszyk, 2018).
pythia?	Verbose pythia!: reports the closing rung and per-rung wall-clock timing. Diagnostic use.	
pythia	Shape-dispatching cascade. Routes to anytime_valid / stats_ineq / prob_simp / oracle by goal shape, then falls through to the @[stat_lemma] aesop ruleset.	2.2. Registry attributes The library uses four attribute-driven registries:
stats.ineq	Discharges concentration and tail-bound inequalities. Dispatches via Mathlib's bound, then positivity, gcongr, linearith, nlinearith.	• @[stat_lemma]: lemma database for the top-level pythia dispatcher. Internally a thin wrapper for @[aesop safe apply (rule_sets := bound pythia)]].
prob_simp	Probability normalization rewrites. simp against the @[prob_simp] set with push_cast/norm_cast fallback. Aliased as pdf_simp.	• @[stats_ineq]: tail-bound and concentration lemmas. Forwards to Mathlib's @[bound] attribute and additionally records the name for the #stats_ineqs introspection command.
anytime_valid	Closes Ville-type maximal inequalities for confidence sequences. Three variants: countable-time, finite-horizon, and explicit-witness.	• @[prob_simp]: probability rewrite rules. Forwards to @[simp] and records the name for #prob_simps.
z3_check	Linear-real-arithmetic Z3 oracle with linearith reconstruction.	• @[anytime_valid_lemma]: Ville-style closers and admissibility lemmas, populating the
cvc5_check	QF_BV / QF_LRA via CVC5 with bv_decide / linearith reconstruction.	Pythia.AnytimeValid aesop ruleset.
vampire_check / e_check	First-order-logic via Vampire / EUSolver aesop reconstruction.	
disprove	Counterexample finder. Asks Z3 for a model of the goal's negation; on sat, fails the proof attempt with a concrete witness. Catches vacuous-truth goal statements.	Efficient attribute, @[cs_family], is a typed-declaration tag rather than a tactic registry: it marks a CSFamily record so it appears in the #cs_families catalogue. User extensions consist of tagging library lemmas with the appropriate attribute. No fork is required.

Table 1. The ten pythia tactics. pythia! is the default entry point for new users. The other nine tactics are exposed for direct invocation when the user knows the goal shape or wants a single rung of the ladder.

anytime_valid. Closes Ville-type maximal-inequality goals for non-negative supermartingales. The dispatch ladder tries direct ville_supermartingale application paths first (countable-time first with exact/refine variants), then the finite-horizon and explicit-witness specialisations, and finally aesop against the named rule-set Pythia.AnytimeValid. Three syntactic variants are exposed: the bare form anytime_valid (countable-time), anytime_valid (horizon := N) (finite-horizon, requiring IsProbabilityMeasure but no integrability), and anytime_valid using h (explicit supermartingale witness). New families register via @[anytime_valid.lemma].

2.3. The Pythia aesop ruleset

The library declares two named aesop rule sets: Pythia (populated by @[stat_lemma]) and Pythia.AnytimeValid (populated by @[anytime_valid.lemma]). Direct invocation is supported via aesop (rule_sets := [Pythia]), equivalent to the inner aesop call inside the pythia dispatcher. This gives users a finer-grained interface when the dispatcher's outer wrapping is not desired.

2.4. Architecture

Why this shape. The two-stage chain mirrors the empirical structure of statistical proofs in our library. Many goals close on a single registered lemma application or a short aesop search at stage 1. The remainder reduce to a linear or arithmetic obligation after rewriting, which the cleanup chain at stage 2 handles. Routing by goal shape is not necessary at the top level. The cost of trying stages in order is small relative to the cost of incorrect early termination.

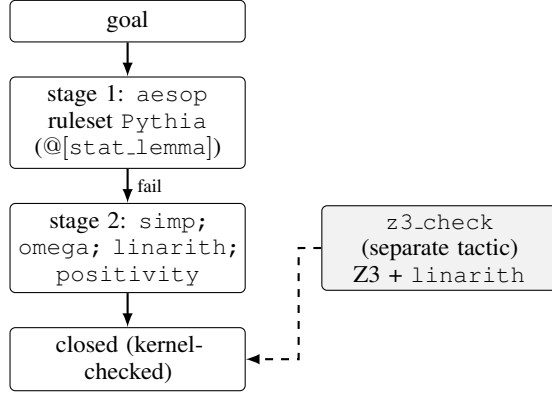


Figure 1. The *pythia* dispatcher chain. Two stages, each kernel-checked independently. Stage 1 is *aesop* against the *Pythia* ruleset (populated by `@[stat_lemma]`). Stage 2 is the cleanup chain. *z3-check* is a separate user-invoked tactic with its own *linarith*-based reconstruction step, not part of the dispatcher chain.

3. Library content

The shipped *pythia* library spans a tactic layer plus a proof-content layer of fifteen applied-math domains. The full manifest at the time of writing carries thirty-two cross-domain theorems with matched Python sim runners (property-based testing, parameter sweeps, mutation testing) plus four hundred and sixty total theorem and lemma declarations across the *Pythia/* source tree, fifty-five of which are tagged `@[stat_lemma]` and registered into the headline cascade. Every theorem listed here is axiom-clean against `{propext, Classical.choice, Quot.sound}` under the Lean 4.28 + Mathlib v4.28.0 toolchain pin. We summarise by group. The appendix lists modules in detail.

3.1. Anytime-valid inference

Howard-Ramdas confidence sequences (Howard et al., 2021), betting-CS (Waudby-Smith & Ramdas, 2024), and the vector-valued extension (Whitehouse et al., 2025) are each backed by a Ville-bound specialization in the *anytime_valid* tactic. The *Pythia.HowardRamdasCS* and *Pythia.BettingCS* modules expose the admissibility certificates of the e-process formalism of Ramdas et al. (2023). The supporting Ville machinery lives in *Pythia.VilleSupermartingale* (countable-time and finite-horizon variants).

3.2. Concentration and tail bounds

stats_ineq dispatches sub-Gaussian and sub-gamma tail bounds, the Bernstein-Bennett MGF bridge, and Hoeffding-style inequalities for bounded random variables.

Modules: *Pythia.SubGaussianMG* (sub-Gaussian martingale concentration), *Pythia.SubGamma* (sub-gamma tail bound), *Pythia.MGFBOundedSubGamma* (the bounded-RV Bernstein-Bennett MGF bound, $\mathbb{E}[\exp(\lambda X)] \leq \exp(\sigma^2 \lambda^2 / (2(1 - b\lambda/3)))$ for centred $|X| \leq b$), and *Pythia.Bernstein* (the iid and martingale Bernstein applications). The MGF bridge composes with *prob.simp* for the integral side and with general Mathlib infrastructure (measurability, integrability) for side conditions.

3.3. E-detectors and merging

The e-process formalism of Ramdas et al. (2023); Shin et al. (2024) yields uniform-time validity by combining e-processes via averaging. *Pythia.EDetector* declares *EProcess* as a non-negative process bounded in expectation under the null and *EDetector* as the first-passage stopping rule with anytime-valid Type-I error $\leq \alpha$. The *combine_eProcesses_avg* lemma (average merge) is the workhorse. Product-merge variants register alongside. The average-merge case is axiom-clean.

3.4. Matrix concentration

Matrix Bernstein (Tropp, 2012) and the Lieb concavity inequality (Lieb, 1973) are the two matrix-side modules. *Pythia.MatrixLieb* states the Lieb concavity step and the operator-monotone log composition. *Pythia.MatrixBernstein* carries the iid matrix-Bernstein head theorem. *Pythia.MatrixBernsteinFull* carries the matrix-martingale extension. The dependent operator-trace inequalities route through Aristotle as oracle (Section 4).

3.5. Cross-domain coverage (15 fields)

The cross-domain manifest covers fifteen fields with Lean proofs and matched Python sim runners. Each entry is a closed-form theorem with property-based-test mutation coverage in the companion runner.

- **Probability and statistics:** the anytime-valid / concentration / matrix sections above plus *Pythia.Risk*, CVaR (Rockafellar & Uryasev, 2000) and *Pythia.Asymptotics*. *DeltaMethod* (scalar) / *DeltaMethodMulti* (multivariate).
- **Information theory:** *Pythia.InfoTheory*. *BretagnolleHuberBinary* (binary-alphabet bound), *Pythia.InfoTheory.DataProcessing* (data-processing inequality), *Pythia.InfoTheory.BinaryEntropy* (Shannon binary entropy).

• Stochastic Pythia.Stochastic.ItoIsometryFiniteDim (finite-dim Itô isometry), Pythia.BDG (Burkholder-Davis-Gundy at $p = 2$), Pythia.MeasureTheory.ConditionalJensen, Pythia.StochasticApproximation.RobbinsSiegmund / RobbinsMonro / Dvoretzky, Pythia.TimeSeries.WoldDecomposition (Wold, 1938), Pythia.TimeSeries.NeweyWest. • Quantum Pythia.Quantum.VonNeumannEntropyNonnegTwoState (qubit von Neumann entropy non-negativity). • Optimal Pythia.OptimalTransport.WassersteinDistanceNonneg (discrete L1 Wasserstein non-negativity). • Game Pythia.GameTheory.MinimaxTwoStrategyBound (von Neumann two-strategy minimax inequality). • Biology: Pythia.Bio.Population (Hardy-Weinberg, Lotka-Volterra equilibria, SIR conservation), Pythia.Bio.MichaelisMentenSaturation (enzyme saturation), Pythia.Bio.MassAction (CRN ODE conservation), Pythia.Bio.Phylogenetics. • Chemistry: Pythia.Chemistry.Arrhenius (rate-law positivity), Pythia.Chemistry.HendersonHasselbalch (pH monotonicity), Pythia.Chemistry.MassActionConservation. • Economics: Pythia.Economics.CobbDouglas (constant returns to scale), Pythia.Economics.CRRRA (utility concavity), Pythia.Economics.CAPM (linear pricing), Pythia.Economics.RiskNeutralCall (non-negativity), Pythia.Economics.Walras (excess-demand identity). • Engineering: Pythia.Engineering.RCTimeConstant, Pythia.Engineering.SignalEnergy, Pythia.Engineering.PowerDissipation. • Mechanical: Pythia.Mechanical.HookeSpring (spring-potential non-negativity). • Control theory: Pythia.Control.Lyapunov (scalar), Pythia.Control.LyapunovDiscrete (discrete-time), Pythia.Control.LyapunovODE. • Operations research / queueing: Pythia.Queueing.ErlangB, Pythia.Queueing.LittlesLaw, Pythia.OR.LittlesLaw.	• calculus: Pythia.Numerical.NewtonQuadraticIterPos, Pythia.Numerical.PicardLindelof, Pythia.Numerical.Lyapunov, Pythia.Numerical.KKT, Pythia.Numerical.Kahan. • Thermodynamics: Pythia.Thermodynamics.CarnotEfficiencyUpperBound. • Mathlib retags + actuarial: Pythia.MathlibTags (AM-GM, Markov, Cauchy-Schwarz with empirical companions), Pythia.Actuarial.{Pareto, Weibull, LogNormal}.
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3.6. Five highlights

We name five representative results that exercise the tactic surface. Module-level statements appear in Appendix A.

1. **Howard-Ramdas Ville bound.** The maximal inequality for the self-normalized confidence sequence (Howard et al., 2021), in `Pythia.HowardRamdasCS`. Closes via `anytime.valid`.
2. **Betting-CS coverage.** The supermartingale coverage certificate of Waudby-Smith & Ramdas (2024), in `Pythia.BettingCS`. Axiom-clean closure.
3. **Bernstein-Bennett MGF bridge.** The bounded-RV MGF bound in `Pythia.MGFBoundedSubGamma`, routing between the variance-aware sub-gamma tail and the Bennett MGF form.
4. **combine.eprocesses.avg.** Average-merge of e-processes (Ramdas et al., 2023; Shin et al., 2024) in `Pythia.EDetector`. Axiom-clean closure with full reconstruction.
5. **Bretagnolle-Huber binary.** The binary-case total-variation lower bound in `Pythia.InfoTheory`. `BretagnolleHuberBinary`. Standard in information-theoretic lower-bound arguments.

Pinning. The library is pinned to Lean 4.28 + Mathlib v4.28.0. The pin is necessary: the modules consume Mathlib’s measure theory and probability subdirectories, which evolve quickly. We track Mathlib release tags rather than master for reproducibility.

4. Aristotle integration as oracle

A subset of statistical proof goals exceeds what `aesop`, `simp`, `linarith`, and the registered `pythia` lemma sets close in practice. We route these to the Aristotle automated theorem prover (Harmonic, 2025) as an oracle. Aristotle returns a

Lean 4 proof script, the script is checked by the Lean kernel inside the calling project, and the result is audited before integration. Aristotle is one of two oracles in the library. The other is Z3 (Section ??).

4.1. Submission pattern

Each Aristotle call is a project tarball submission. The flow is: state the theorem in Lean with a placeholder, package the project without the Lean build cache (`.lake` excluded), submit via the Aristotle endpoint, monitor for completion, integrate the returned proof, and run the axiom audit. The audit is the verification step that distinguishes a kernel-checked Aristotle proof from a vacuous closure.

4.2. Vacuous-truth refusal

ATPs can close a theorem by deriving a contradiction from its hypotheses, then concluding anything. The kernel accepts such proofs. The proof is sound. The theorem statement is the bug.

Case (2026-04-26). We submitted a candidate `wald_wolfowitz_optimal` theorem. Aristotle returned a closure that combined a Bool partition with a log-boundary constraint to force both $\alpha \geq 1$ and $\alpha < 1$. The kernel accepted the proof. The audit flagged the hypothesis chain as self-contradictory. We rejected the closure, restated the theorem in two-measure form (separating the null and alternative measures explicitly), and resubmitted. The two-measure form was the intended target. The original Bool-partition form was a statement bug.

Audit policy. The library treats vacuous closures as theorem-statement bugs, not as prover successes. Concretely: every Aristotle-returned proof is audited for contradictory premises before integration. Closures flagged as vacuous block the corresponding theorem from the axiom-clean library until the statement is reformed. This policy is not unique to Aristotle. The same audit applies to any external prover, including human proofs that close via a hypothesis contradiction the author did not notice.

4.3. Z3 as a second oracle

The `z3.check` tactic dispatches linear-real-arithmetic goals to Z3 (de Moura & Bjørner, 2008). Z3 returns a verdict (`unsat` of the negated goal). The verdict is then reconstructed inside Lean by `linarith`. The Lean kernel is the final word. The pattern is the CoqHammer (Czajka & Kaliszyk, 2018) reconstruction strategy adapted to Lean 4. Z3 cannot smuggle axioms. `linarith` is axiom-clean and the reconstruction step certifies the witness inside the kernel. If `linarith` cannot reconstruct, the `z3.check` call fails loud rather than silently trusting Z3.

4.4. When to call Aristotle

The library convention is to attempt local closure first. If the goal closes by `rfl`, `simp`, `positivity`, `linarith`, `nlinarith`, `polyrith`, or the registered `stat_lemma` database, no Aristotle call is made. Aristotle is reserved for hard structural Mathlib gaps: measure-theoretic σ -additivity steps, dominated-convergence side conditions, matrix-concentration intermediate lemmas, and similar obligations for which a hand-Mathlib closure either does not exist or would require an order of magnitude more dispatch. The vacuous-closure audit is the gate on integration. The dispatch threshold is only the call decision.

5. Comparison and discussion

5.1. Comparison to existing automation

aesop. Aesop (Limperg & From, 2023) is the workhorse forward-search tactic in Lean 4. `pythia` declares two named aesop rule sets (`Pythia` and `Pythia.AnytimeValid`) and uses aesop in stage 1 of the dispatcher chain. `pythia` is not a replacement for aesop. It is a statistics-domain registry plus a Z3 oracle plus a vacuous-closure audit, layered on top of aesop.

Sledgehammer. Sledgehammer (Paulson & Blanchette, 2010; Blanchette et al., 2011) dispatches Isabelle/HOL goals to external ATPs (Vampire, E, Z3, CVC4) and reconstructs the proof. There is no direct Lean 4 equivalent. `pythia` addresses the statistics-domain case via `z3.check` for linear-real-arithmetic and Aristotle as a structural oracle. A general-purpose Lean Sledgehammer is out of scope for this paper.

CoqHammer. CoqHammer (Czajka & Kaliszyk, 2018) dispatches Coq goals to external ATPs with kernel-checked reconstruction. `z3.check` in `pythia` adopts the same reconstruction template (witness from external prover, in-kernel checking via `linarith`). CoqHammer is the design template. We do not claim a Lean-side reimplement of CoqHammer’s full premise selection.

Mathlib’s existing tactic suite. Mathlib supplies `linarith`, `polyrith`, `nlinarith`, `positivity`, `measurability`, `bound`, `gcongr`. None of these is a statistics-domain hammer. `stats.ineq` is the closest `pythia` relative of the Mathlib bound family: the `@[stats.ineq]` attribute re-tags lemmas as `@[bound]`, so `bound`’s mature dispatch picks them up automatically. `pythia` composes with all of the listed Mathlib tactics. The fall-through chain at stage 2 of the dispatcher combines `simp`, `omega`, `linarith`, and `positivity`. The chain is the explicit composition.

5.2. Aristotle vs DSPv2 vs Opus 4.6 (illustrative)

Table 2 reports four FormalAVS problems (Anonymous, 2026) on which Aristotle, DSPv2 (DSPv2-7B GPTQ-int8, a quantised local-GPU prover), and Opus 4.6 (an API generalist drafter) were each given a single-attempt protocol (Aristotle is one session per problem. DSPv2 and Opus run pass@5 at $N = 5$ on the 48-slate.) The full benchmark numbers are in the dataset paper. The point of this subset is to expose the four solver regimes that the benchmark tail surfaces. Per-problem solve outcomes are kernel-checked.

Problem	Aristotle	DSPv2	Opus 4.6
betting_is_kelly_optimal*	1/1	n/a	n/a
etavector.eq.sqrt.two.mul.etahr	1/1	1/5	0/5
c.sharp_ranking	1/1	0/5	0/5
equivalence_break_at_finite_precision	0/3 [†]	0/5	0/5

Table 2. Illustrative four-problem subset of FormalAVS. Each cell is (closures)/(attempts). Aristotle is one session per problem. DSPv2 and Opus are pass@5 at $N=5$. The four rows expose four distinct solver regimes: universal closure (top), majority closure (etavector, the $\sqrt{2}$ identity Kimi and Mistral also close at $N=5$), Aristotle-monopoly (c.sharp_ranking), and solver-unreached frontier (equivalence_break). *Companion-archive entry, not in the 48-slate drafter sweep, included for the universal regime. [†]Aristotle returned a counterexample on the as-stated form (missing symmetry hypothesis). The corrected form remains open. Full per-tier benchmark numbers in Anonymous (2026).

The wider table on the full benchmark subset is in Appendix C.

5.3. Limitations

Three concrete gaps. (i) Domain scope: pythia is statistics-domain only. It does not address general proof automation in Lean 4. (ii) Toolchain pin: the library ships pinned to Lean 4.28 + Mathlib v4.28.0. Mathlib evolves quickly. Each major Mathlib release typically requires a porting pass. (iii) Aristotle dependency: the hardest closures route to a proprietary external prover. The z3_check oracle is open-source. The Aristotle oracle is not. The library is usable without Aristotle for any goal that closes via the pythia dispatcher chain or z3_check. A subset of structural Mathlib gaps requires either Aristotle or a hand proof.

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A. Library inventory

The shipped library at the AI4MATH 2026 submission cut spans the modules below. Each line lists module path and one-line summary. All listed modules are axiom-clean against $\mathcal{A}_{\text{core}} = \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$ unless explicitly flagged as scaffold.

A.1. Anytime-valid inference

- `Pythia.HowardRamdasCS`: Howard-Ramdas Ville bound for self-normalized confidence sequences (rate $\eta_{\text{HR}}(b) = \sqrt{b \log 2}$). Axiom-clean.
- `Pythia.BettingCS`: betting-CS coverage certificate (rate $\eta_{\text{betting}}(b) = 1/\sqrt{b \log 2 + 1}$). Axiom-clean.
- `Pythia.BettingStrategy`: betting-strategy admissibility scaffolding for the `anytime.valid` dispatch.
- `Pythia.VilleSupermartingale`: countable-time and finite-horizon Ville inequalities. The `ville_supermartingale`, `ville_ineq`, and `ville_supermartingale.finite` lemmas are the workhorse.
- `Pythia.EDetector`: e-process formalism, with `combine_eprocesses_avg` (average-merge) as the workhorse closer. Anytime-valid Type-I error control via Markov-at-stopping.

A.2. Concentration

- `Pythia.SubGaussianMG`: sub-Gaussian martingale concentration. The Chernoff exponent $\exp(-\tau^2/(2\sigma^2 N))$ bound. Axiom-clean.

- `Pythia.SubGamma`: sub-gamma tail bound. Axiom-clean.

- `Pythia.MGFBoundedSubGamma`: bounded-RV Bernstein-Bennett MGF bound, $\mathbb{E}[\exp(\lambda X)] \leq \exp(\sigma^2 \lambda^2 / (2(1 - b\lambda/3)))$ for centred $|X| \leq b$. Axiom-clean.

- `Pythia.Bernstein`: iid and martingale Bernstein applications.

A.3. Matrix concentration

- `Pythia.MatrixLieb`: Lieb concavity step (Lieb, 1973) and operator-monotone log composition. Scaffold. Dependent operator-trace inequalities route to Aristotle.
- `Pythia.MatrixBernstein`: iid matrix-Bernstein head theorem (Tropp, 2012). Scaffold.
- `Pythia.MatrixBernsteinFull`: martingale extension. Scaffold.

A.4. Other modules

- `Pythia.Control.LyapunovDiscrete`: discrete-time Lyapunov stability for linear systems.
- `Pythia.Queueing.ErlangB`: Erlang-B blocking formula.
- `Pythia.Queueing.LittlesLaw`: Little’s law steady-state identity.
- `Pythia.TimeSeries.WoldDecomposition` (Wold, 1938).
- `Pythia.Risk.CVaR` (Rockafellar & Uryasev, 2000): the Rockafellar-Uryasev variational characterisation of CVaR.
- `Pythia.Asymptotics.DeltaMethod`, `Pythia.Asymptotics.DeltaMethodMulti`: scalar and multivariate delta method.
- `Pythia.InfoTheory.BretagnolleHuberBinary`: binary-case total-variation bound via the Bhattacharyya bridge.
- `Pythia.MeasureTheory.OptionalStoppingUnbounded`, `Pythia.MeasureTheory.PathMeasureRN`: structural support for the Ville chain.

B. Worked examples

Each example below is a literal regression test from the `pythia` test suite. Every example compiles axiom-clean against $\mathcal{A}_{\text{core}}$.

B.1. anytime_valid: Ville-bound closer

The countable-time variant closes the canonical Ville maximal inequality on a non-negative supermartingale.

```
import Pythia.Tactic.AnytimeValid
open MeasureTheory ProbabilityTheory
  ENNReal Pythia

example {Omega : Type*} {m0 :
  MeasurableSpace Omega}
{mu : Measure Omega} [IsFiniteMeasure mu]
  ]
{f : Nat -> Omega -> Real}
{F : Filtration Nat m0}
(hsup : Supermartingale f F mu)
(hnn : forall t omega, 0 <= f t omega)
(hint : Integrable (f 0) mu)
{c : Real} (hc : 0 < c) :
mu {omega | exists t : Nat, f t omega >=
  c}
  <= (Integral.integral mu (f 0)).
    toNNReal / c.toNNReal := by
  anytime_valid
```

The tactic dispatches via the `ville.supermartingale` exact path. Hypotheses are discharged by assumption. No registered rule is consulted. The kernel-checked term reduces under `{propxt, Classical.choice, Quot.sound}`.

B.2. z3_check: linear-real-arithmetic with reconstruction

A three-step linear chain over $\mathbb{R}$. Z3 is queried as a ranking filter. `linarith` produces the actual proof term.

```
import Pythia.Tactic.Z3Check
open Pythia

example {a b c : Real} (h1 : a < b) (h2 : b
  <= c) :
  a < c := by
  z3_check
```

Trace: the encoder produces the SMT-LIB query `(set-logic QF_LRA) ... (assert (< v_a v_b)) (assert (<= v_b v_c)) (assert (not (< v_a v_c))) (check-sat)`, which Z3 returns `unsat`. `linarith` then constructs the Farkas certificate inside Lean. Z3 cannot smuggle axioms. If `linarith` disagreed, the tactic would fail loud.

B.3. pythia: chain-rule fall-through to aesop

A goal with no matching `@[stat_lemma]` but a closed-form chain in `linarith`. Stage 1 (aesop with the Pythia ruleset) declines. Stage 2 (the cleanup chain combining `simp`, `omega`, `linarith`, and `positivity`) closes.

```
import Pythia.Tactic.Pythia
open Pythia
```

```
example (a b c d : Real)
  (h1 : a <= b) (h2 : b <= c) (h3 : c <= d)
  ) :
  a <= d := by
  pythia
```

The dispatcher's first-combinator tries `aesop`, observes no applicable `@[stat_lemma]` closer, then tries the cleanup chain. `linarith` closes the chain in one step.

B.4. Registry-extend: tagging a user theorem

User extensions consist of tagging library lemmas with the appropriate attribute. Below: a trivial real-arithmetic identity registered for `pythia` via `@[stat_lemma]`.

```
import Pythia.Tactic.Pythia
open Pythia

@[stat_lemma]
theorem add_zero_real (x : Real) : x + 0 =
  x := by ring

-- pythia now closes goals matching the
  registered lemma:
example (x y : Real) : (x + 0) + (y + 0) =
  x + y := by
  pythia
```

The attribute re-elaborates as `@[aesop safe apply (rule_sets := [Pythia])]`. The lemma then joins the Pythia aesop ruleset and is consulted by `pythia` on every subsequent invocation. No fork or rebuild of the tactic implementation is required.

C. Aristotle vs DSPv2 vs Opus 4.6 (wider subset)

Table 3 reports a wider subset of FormalAVS (Anonymous, 2026) per-tier closure rates for the three solver families. Aristotle is one-attempt-per-target. DSPv2 and Opus are `pass@5` at $N=5$ on the 48-target slate (T0 through T2 plus T3). Cells are *(closures)/(targets)* at the tier denominator, with T4 (library-gap) and T5 (capability-ceiling) appended.

Per-target capability-wall tail. Table 4 reports the per-target closure matrix on the five capability-wall tail problems used in the dataset paper to classify solver regimes.

Solver	T0	T1	T2	T3	T4	T5
Aristotle	7/7	33/34	7/8	6/6	0/4	0/1
DSPv2-7B GPTQ-int8	3/7	8/34	0/8	1/6	0/4	0/1
Opus 4.6	0/7	13/34	0/8	0/6	0/4	0/1

Table 3. Wider per-tier comparison from FormalAVS. Aristotle: one attempt per target. DSPv2 and Opus: pass@5 at $N=5$. T4 is solver-hit-Mathlib-gap (every solver 0/4 for the same library-gap reason). T5 is the all-solver capability-ceiling target. The 6/6 vs 1/6 vs 0/6 column at T3 (cross-function inequality wall) is the benchmark’s cleanest solver-specific signal. The single DSPv2 T3 closure is `etavector.eq.sqrt.two.mul.eta` (the $\sqrt{2}$ identity, also closed by Kimi K2.5 and Mistral Large 3 in the wider drafter sweep). Source: dataset paper (Anonymous, 2026), Tables 1-2 reproduced verbatim.

Target	Aristotle	Kimi K2.5	Sonnet 4.6	Gemini 3 Pro
betting_is_kelly_optimal*	1/1	1/1	1/1	1/1
c_vector_eq_sqrt_two_mul_c_HR	1/1	1/1	0/5	1/2
c_sharp_ranking	1/1	0/9	0/5	1/3
quantized_entropy_bound	1/1	0/9	0/5	0/3
equivalence_break_at_finite_precision	0/3 [†]	0/19	0/15	0/13

Table 4. Per-target capability-wall tail. Cells are $(\text{closures})/(\text{attempts})$. Five distinct solver regimes appear: universal closure, majority closure, drafter-monopoly (Gemini’s `gcongr-prior` closes `c_sharp_ranking`), Aristotle-monopoly, and solver-unreached. Source: Anonymous (2026), Table 4. *Companion-archive entry, not in the 48-slate drafter sweep. [†]Aristotle: two sessions timed out at the 20-minute compute gate. One returned a counterexample on the as-stated form (missing symmetry hypothesis, (Anonymous, 2026) §Aristotle-refute).

D. Reproduction

Toolchain pin. Lean 4.28 + Mathlib v4.28.0. The repository’s `lean-toolchain` file declares the exact toolchain. The Mathlib dependency in `lakefile.toml` pins the v4.28.0 release tag.

Build. The library URL is redacted for double-blind review (full URL in camera-ready). Reproduction commands once the URL is supplied:

```
git clone <pythia-url-redacted>
cd pythia
lake exe cache get
lake build
```

Axiom audit. Each shipped module ends with a `#print axioms` block. The expected audit set is `{propext, Classical.choice, Quot.sound}`. Any deviation fails the audit.

FormalAVS benchmark. The full benchmark is reported in Anonymous (2026). Reproduction commands for the benchmark are in that paper’s reproduction appendix. The benchmark URL is redacted for double-blind review.

E. Related work, extended

E.1. Lean 4 tactic infrastructure

Mathlib ships a tactic suite the `pythia` dispatcher composes with. `linarith` produces Farkas certificates, `polyrith` routes polynomial-ideal goals via Sage with kernel-checked reconstruction, `nlinarith` extends `linarith` to a small non-linear fragment, `positivity` discharges sign goals, `measurability` closes σ -algebra-membership obligations, `gcongr` discharges generalised-congruence monotonicity, and `bound` dispatches monotone-composition goals via an attribute-driven rule database. The closest Mathlib relative to `pythia` is `bound`: the `@[stats_ineq]` attribute literally re-tags lemmas as `@[bound]`, so the underlying `bound` dispatch picks them up automatically. Aesop (Limpert & From, 2023) supplies the `rule_sets` mechanism the `pythia` dispatcher uses. `pythia` is therefore a thin domain-registration layer over `aesop` plus a Z3 oracle.

E.2. Sledgehammer (Isabelle)

Sledgehammer (Paulson & Blanchette, 2010; Blanchette et al., 2011) dispatches Isabelle/HOL goals to Vampire, E, Z3, CVC4, with proof reconstruction inside Isabelle’s kernel via Metis or `smt-tactic` replay. The reconstruction discipline is the template for both CoqHammer and `z3_check`. There is no Lean 4 equivalent at the time of writing. `pythia`

does not attempt the Isabelle Sledgehammer’s full premise-selection front end. The library ships only the linear-real-arithmetic instance via `z3_check`.

E.3. CoqHammer (Coq)

CoqHammer (Czajka & Kaliszyk, 2018) dispatches Coq goals to Vampire, E, Z3, CVC4 with kernel-checked reconstruction. Premise selection in CoqHammer is k-NN over a featured representation of the Coq context. The reconstruction step is a small certifying tactic (`sauto`) that replays the ATP’s witness inside the Coq kernel. `z3_check`’s reconstruction-via-`linarith` mirrors this template at small scale: external prover supplies the verdict, in-kernel tactic supplies the proof term, kernel is the final arbiter. We do not claim a full Lean-side reimplementa-tion of CoqHammer’s premise-selection layer.

E.4. Statistical formalisation libraries

Mathlib’s `Probability.Martingale` module supplies Doob decomposition and Azuma’s inequality but does not instantiate the canonical confidence-sequence families, the per-family rate functions, or Ville’s inequality for the betting construction. The `pythia` library closes that gap. The FormalAVS benchmark (Anonymous, 2026) measures the result. Coq’s `mathcomp.analysis` and the Isabelle/HOL probability formalisations cover overlapping but disjoint sub-areas, and are not attribute-driven hammers.